Chapter 6 lecture script: Fitting ecological response curves

Draft for the course author's review. **Target: 150 minutes in total, with 121 minutes of slides on 32 slides (including the discussion, the three practice questions and one live demo of 18 minutes), 9 minutes of slack and two breaks of 10 minutes.** The breaks separate the three blocks of the lecture. The only code that is run live is the demo of slide 23, from the script `demo-cycle.R`.

This script follows the Chapter 6 theory from the introduction to the recipe, with a short introduction to the R practical. The lecture goes round the modelling cycle once with one data set, the reed frog functional response: the model, the fit, the uncertainty of the parameters and of the curve, the check with residuals, the revision of the observation model and the comparison of candidate models. The slides carry keywords only; the narrative is here.

Slide 3 uses your diagram `ModelCycle.png` from the Chapter 1 lecture folder. Slides 5 (`curve-distribution.png`), 11 (`fitted-curve.png`), 14 (`parameter-derivatives.png`), 17 (`curve-band.png`) and 21 (`residuals-binomial.png`) use figures in `lecture-assets/`. All five are unchanged copies of chapter figures: `fig-curve_distribution`, `fig-fitted_curve`, `fig-curve_band` and `fig-residuals_binomial` of Chapter 6, and `fig-parameter_derivatives` of Chapter 5, because the Chapter 6 theory has no figure of partial derivatives. Their code is in `lecture-assets/figures.R`, which runs the chapter's chunks in the chapter's order with the chapter's seed, so the residual plot is the one on the chapter page. If the chapter figures change, run that script again (it takes a few seconds).

Timing and slide sequence

Times are measured from the start of the session and include the breaks.

Slide	Topic	Minutes	Elapsed time
1	Ecological Models and Data Analysis	1	0–1
2	Overview of the lecture	2	1–3
3	Where we are: the modelling cycle	1	3–4
4	Two pillars: functions and distributions	3	4–7
5	A response curve as a parameter of a distribution	4	7–11
6	What the curve must return	4	11–15
7	Practice 1: seedlings and light	3	15–18
8	The likelihood of a response curve	4	18–22
9	The negative log-likelihood in R	4	22–26
10	Starting values and the fit	4	26–30
11	Reading the fitted curve	3	30–33
12	The gradient and automatic differentiation	3	33–36

Slide	Topic	Minutes	Elapsed time
	Break	10	36–46
13	Precision and correlation of curve parameters	4	46–50
14	Where the data lie	4	50–54
15	Likelihood profile intervals of curve parameters	4	54–58
16	The uncertainty of a fitted curve	4	58–62
17	A confidence band	4	62–66
18	Practice 2: handling time of a ladybird	3	66–69
19	Simulating from a fitted model	3	69–72
20	Scaled quantile residuals	4	72–76
21	Reading residual plots	4	76–80
22	Discussion: what does a residual pattern tell us?	3	80–83
	Break	10	83–93
23	Live demo: the modelling cycle for one model	2 + 16	93–111
24	Revising the observation model	4	111–115
25	Practice 3: fruit set in inflorescences	3	115–118
26	Candidate models	3	118–121
27	AIC, AICc and BIC	5	121–126
28	Leave-one-out cross-validation	4	126–130
29	Comparing the candidates	4	130–134
30	A recipe for a response curve	2	134–136
31	The Chapter 6 R practical	2	136–138
32	Summary and closing questions	3	138–141
	Slack for discussion	9	141–150
	Total	**150**	

Slides 121 minutes (blocks of 36, 37 and 48, the demo included), slack 9, breaks 20: 150 minutes. The break before the demo lets you open R and run section 0 of `demo-cycle.R`.

Slides 1 and 2 — Ecological Models and Data Analysis; Overview of the lecture

In this lecture, we learn how to fit an ecological response curve to data: a deterministic function of a predictor describes one parameter of the distribution of each observation, and the parameters of the function are estimated by maximum likelihood. Once the curve is fitted, we ask how precisely

its parameters and the curve itself are estimated, whether the fitted model describes the data, how to revise it where it does not, and which of several candidate models the data support.

The session has three blocks, separated by two breaks. The first block builds the model and fits it, and ends with the gradient that RTMB supplies to the optimiser. The second is about the uncertainty of the parameters and of the curve, and about checking the fitted model with residuals calculated by simulation. The third starts with a live demo of the whole cycle for one model, then revises the observation model and compares four candidate models with information criteria and cross-validation. It ends with the recipe, the practical and the summary.

Delivery

Read the overview at the pace of the list, one keyword and its question at a time, and do not explain any term yet.

Slide 3 — Where we are: the modelling cycle

This lecture goes round the whole cycle once, with one data set. In the model box it is the entry "covariates (5 and 6)": Chapter 5 studied the deterministic functions, and this chapter makes one of them a parameter of a distribution. In the comparison box it adds BIC and cross-validation to the AIC of Chapter 4, and it checks how well a fitted model describes the data. A check that fails sends us back to the model, which is why modelling is a cycle, and the lecture follows this path: model, estimation, check, revision, comparison, answer.

Delivery

One minute. Point to "covariates (5 and 6)", "information criteria (4 and 6)" and "cross-validation (6)", and to the arrow that leads from comparison back to the model.

Slide 4 — Two pillars: functions and distributions

The book has been built on two pillars. Chapters 2 to 4 gave us probability distributions for the observations and estimated their parameters by maximum likelihood; in Chapter 4 every observation had the same distribution. Chapter 5 studied deterministic functions that describe how an expected response changes with a predictor, without any statistics. A response curve joins the two: the curve gives a parameter of the distribution of each observation, so the parameter changes from one observation to the next with the predictor.

One example runs through the lecture: the functional response of dragonfly larvae that prey on reed frog tadpoles, from Vonesh and Bolker (2005), "Compensatory larval responses shift trade-offs associated with predator-induced hatching plasticity", *Ecology* 86: 1580–1591. The data are `ReedfrogFuncresp` in the package `emdbook`. Sixteen tanks were stocked with 5 to 100 tadpoles together with three dragonfly larvae, and the number of tadpoles killed in 14 days was counted. These are the data that Chapter 5 used as a backdrop: it read an asymptote of about 60 tadpoles killed and a half-maximum of about 120 tadpoles per tank by eye, and those values become our starting values on slide 10. We treat every tank as an independent observation; Chapter 7 extends the models to observations that come in groups.

Delivery

The two columns appear one at a time, then the example and the takeaway. Ask students what the response and the predictor are in these data before the example appears (the number killed; the initial number of tadpoles).

Slide 5 — A response curve as a parameter of a distribution

A **response curve** gives one parameter of the distribution of each observation as a function of a predictor. Let Y_i be the i-th observation and x_i the value of the predictor at which it was made. The parameter of its distribution is $\theta_i = f(x_i; \boldsymbol{\beta})$, where f is a deterministic function of the kind studied in Chapter 5 and $\boldsymbol{\beta}$ is the vector of its parameters (the semicolon separates the input of the function from its parameters). Usually θ_i is the mean, because the curves of Chapter 5 describe how an expected response changes with a predictor. The other parameters of the distribution, such as the standard deviation of a Normal model or the shape of a Negative Binomial model, are often shared by all the observations.

So every observation has a distribution of the same family, with its own value of the parameter that follows the curve. We still treat the observations as independent, but they are no longer identically distributed: two observations made at different values of the predictor have different distributions, and the distribution of each depends on the known value of its predictor. This is the change from Chapter 4. The observation model is a choice, in the same way as the curve is (Chapter 2, "Choosing a distribution"), and in this lecture we will revise it (slide 24).

For the reed frogs, K_i is the number killed in tank i, which was stocked with N_i tadpoles. Each tadpole of a tank is either killed or not, so we use a Binomial model with N_i trials, in which the curve gives the probability that a tadpole is killed, $p(N) = \alpha/(1 + \alpha H N)$. Here α is the attack rate and H the handling time of the Holling type II form of the disc equation (Chapter 5, "Reparameterisation"). The predictor, the initial number of tadpoles, is also the number of trials.

Figure walkthrough

The blue points are the 16 tanks. The vermillion line is the expected number killed under the curve with the values read by eye in Chapter 5, $\alpha = 0.5$ per experiment and $H = 1/60$ experiments per tadpole killed. At 10, 50 and 100 tadpoles per tank, the horizontal bars show the Binomial probability of each possible number killed, with a length proportional to the probability. They are the distributions of three tanks: each tank has its own, centred on the curve, and the spread increases with the number of tadpoles in the tank. The figure shows the model before any fitting; the next slides fit it.

Slide 6 — What the curve must return

The curve must return a value in the range of the parameter that it describes. A mean count cannot be negative and a probability must lie between 0 and 1, whereas the functions of Chapter 5 are not restricted in this way for every value of their parameters and of x. The table gives the usual cases. For a Normal mean any curve will do. For the mean of a Poisson or Negative Binomial distribution we need a curve that is positive for every x, or the exponential of another curve. For a Binomial probability we need a curve between 0 and 1 for every x, or the inverse logit of

another curve. Chapter 5 ("Range and composition") showed how an outer function gives a curve the range that a quantity requires.

For the reed frogs the curve describes the probability and not the number killed, because a Binomial distribution is written in terms of the number of trials and a probability. The expected number killed is then $N\,p(N) = \alpha N/(1 + \alpha H N)$, which is the disc equation of Chapter 5. Under the assumptions of the disc equation, the probability of being killed decreases with density because the larvae spend more of their time handling the prey that they have caught. The units follow from Chapter 5: the tadpoles killed were counted over the whole experiment, so α is a fraction per experiment (at low density, a tadpole is killed with probability close to α), and H is in experiments per tadpole killed. With $0 < \alpha < 1$ and $H > 0$, $p(N)$ lies between 0 and α for every density, so this curve has the range that the Binomial distribution requires without an outer function. When we fit it, we keep α between 0 and 1 and H positive with the transformations of Chapter 4 (slide 9).

The takeaway is the general point: the range of the curve is part of the model, and it is chosen together with the observation model.

Transition

The next slide asks students to apply this to another distribution.

Slide 7 — Practice 1: seedlings and light

Short activity — allow about two minutes of work and one to discuss

Students work on paper. Expected answers, from the chapter's solution:

1. The curve describes the mean λ of the Poisson distribution, the expected number of seedlings per quadrat, which must be positive.
2. No: the straight line becomes negative for some values of x unless a and b are restricted. Its exponential, e^{a+bx}, is positive for any a, b and x.
3. Not if some quadrats have no light. The Michaelis–Menten function is positive for $x > 0$, but it is zero at $x = 0$. A Poisson distribution with a mean of zero gives probability one to a count of zero, so a quadrat without light that contains a seedling would have a probability of zero under the model, and the log-likelihood could not be calculated.

The third answer is the one that students usually miss: a curve can have the right range almost everywhere and still fail at a value of the predictor that occurs in the data.

Slide 8 — The likelihood of a response curve

The likelihood of a response curve is calculated as in Chapter 4: the observations are independent, so the log-likelihood is the sum of the log probabilities (or densities) of the observations. The difference is that each probability is calculated with the parameter of its own distribution, which the curve gives at the predictor value of that observation. In the equation, y_i is the observed value of Y_i and $P(Y_i = y_i \mid \beta)$ is the probability that the model assigns to it when the parameter of its distribution is $f(x_i; \beta)$; any shared parameter of the distribution is added to β.

Everything else is the same as in Chapter 4: we write the negative log-likelihood (NLL) with two lists, transform the constrained parameters, connect the NLL to the data with `cmb()`, process it with `MakeADFun()`, minimise it with `optim()` and check the convergence. Two things are new. The list of data holds the predictor as well as the response, and the starting values of the parameters of the curve are read from a plot of the data, as in "Starting values by eye" in Chapter 5.

For the reed frogs, each term of the sum is the log of the Binomial probability of the observed number killed in one tank, with N_i trials and the probability $p(N_i)$ of that tank, and the log-likelihood adds up the 16 terms. It is a function of α and H only, because the numbers of tadpoles and the numbers killed are fixed by the data.

Slide 9 — The negative log-likelihood in R

We load RTMB and RTMBdist (in that order, as in every chapter since Chapter 2) and the data, and put the two variables that the NLL uses in the list of data: `initial`, the number of tadpoles at the start of the experiment in each tank (the predictor and the number of trials), and `killed`, the number killed (the response).

The NLL has the form of Chapter 4. `getAll()` makes the elements of both lists available by name. The attack rate is a probability, so the optimiser works with its logit, `logit_alpha`, which `plogis()` turns back into α; the handling time is positive, so the optimiser works with its logarithm, `log_H`, which `exp()` turns back into H. Because `initial` is a vector, the line of the curve returns a vector with one probability for each tank, and `dbinom()` uses the number of trials and the probability of each tank. The sum of the log probabilities, with a minus sign, is the NLL.

Delivery

Show the chunk without running it; the demo of slide 23 runs it. Point at the line of the curve and say that this is the only line that a response curve adds to an NLL of Chapter 4: the rest is the same recipe.

Slide 10 — Starting values and the fit

Chapter 5 read $a = 60$ tadpoles killed and $b = 120$ tadpoles per tank from a plot of these data, in the Michaelis–Menten form $ax/(b + x)$. They correspond to $\alpha = a/b = 0.5$ per experiment and $H = 1/a = 1/60$ experiments per tadpole killed. We give them to `MakeADFun()` on the transformed scales, $\text{logit}\,(0.5) = 0$ and $\log(1/60)$, connect the NLL to the data with `cmb()`, and minimise it with `optim()`, BFGS and the gradient `gr` (slide 12 says what the gradient is). A convergence code of zero is the first check. The estimates are then transformed back to the original scale: $\hat{\alpha} \approx 0.526$ per experiment and $\hat{H} \approx 0.0166$ experiments per tadpole killed.

Starting values read from a plot are the general route for a response curve: the curve has parameters with a meaning (an initial slope, an asymptote, a half-maximum), and Chapter 5 taught how to read them from the data. Any starting values from which the optimiser reaches the maximum will do; values by eye are a reliable way to find such values.

Delivery

Show the chunk without running it. The two estimates are in the comments.

Slide 11 — Reading the fitted curve

A figure of a fitted response curve shows the observations as points and the expected response under the fitted model as a line. By invariance (Chapter 4), the estimates of other quantities of the curve follow from the estimates of its parameters. Two are useful here: the largest number of tadpoles that the larvae could kill in one experiment, $1/H$, which is the asymptote of the expected number killed, and the density at which the expected number killed reaches half of that asymptote, $1/(\alpha H)$. These are the parameters a and b of the Michaelis–Menten form.

With the estimates, the larvae could kill at most about 60 tadpoles in one experiment, and the expected number killed reaches half of this at about 115 tadpoles per tank. These values are close to the 60 and 120 that Chapter 5 read by eye. That is not surprising: the initial slope of the curve can be read from the tanks at low density, and the asymptote that Chapter 5 chose happened to be close to the estimate. Chapter 5 also found that these data say little about the asymptote, and slides 13 to 15 show what this means for the uncertainty of the estimates.

A convergence code of zero is the first check of a fit. The next is whether the fitted model describes the data, which we check with residuals in slides 19 to 21. When the code is not zero, the note "When a fit fails" in Chapter 4 says what to try.

Figure walkthrough

The blue points are the 16 tanks. The solid vermillion curve is the expected number killed under the Binomial model fitted by maximum likelihood, and the dashed curve the one with the values read by eye. The two nearly coincide over the range of the data, and both are still rising at 100 tadpoles per tank. The figure does not show how uncertain the fitted curve is; that is slide 17.

Slide 12 — The gradient and automatic differentiation

Since Chapter 4 we have given every NLL to `MakeADFun()` before fitting it, and we said that RTMB supplies the slope that BFGS uses. We can now say what that slope is. The **gradient** of the NLL is the vector of its partial derivatives with respect to the parameters (Chapter 5, "Derivatives with respect to parameters"), that is, its slope in the direction of each parameter. RTMB calculates it by **automatic differentiation**: from the code of the NLL, exactly up to rounding error, and without a formula from us. The element `gr` of the object returns the gradient at the parameter values that we give it, on the scale that is optimised, with one column per parameter.

At the starting values the gradient is -4.30 for `logit_alpha` and 3.20 for `log_H`. Its signs tell BFGS in which direction to move each parameter to lower the NLL: a negative slope means that increasing the parameter lowers the NLL, so α should go up; a positive one that H should go down. The estimates moved that way, slightly. At the estimates both elements are close to zero, as they should be at a minimum.

An exact gradient matters for response curves for three reasons. First, BFGS uses it at every step. Without `gr`, `optim()` approximates the gradient by evaluating the NLL at nearby values of every parameter (Chapter 8 explains how), which costs time and accuracy. Second, the parameters of a curve are often correlated (slide 13), so the NLL forms a narrow and tilted valley like the one of the owl model in Chapter 4. In such a valley the direction of the next step depends on small differences

between the partial derivatives, and an inexact gradient can stop the search before it reaches the minimum. Third, the same derivatives are behind the other tools of the course: `sdreport()` calculates the curvature of the NLL from them, `ADREPORT()` uses them for the standard errors of quantities calculated from the parameters (slide 16), and in Chapter 7 they are needed to integrate random effects out of the likelihood.

For RTMB to calculate these derivatives, the NLL must be written with operations and functions that RTMB knows, which is why the probability functions of the course come from RTMB and RTMBdist. RTMB works out the derivatives from the calculations that the NLL carried out when `MakeADFun()` ran it at the starting values, so every calculation that involves a parameter must have a derivative. A branch that depends on a parameter (an `if` that compares a parameter with a value) does not: if it were recorded, only the side taken at the starting values would be used afterwards. RTMB therefore refuses such a comparison, and `MakeADFun()` stops with the error "Comparison is generally unsafe for AD types". In the same way, a curve with a sharp change at an unknown threshold has no derivative at the change, and a smooth curve that approximates the sharp change is the usual solution. The practical advice of the chapter: call the NLL directly with two lists on sensible parameter values first, and inspect `fn` and `gr` at the starting values before fitting.

Optional remark (the box of the chapter)

The technical detail box says how automatic differentiation works. `MakeADFun()` runs the NLL once and records every elementary operation that it carries out on the parameters (each sum, product, division, exponential and logarithm) in order. This record is called a **tape**. The derivative of each operation is known, and the chain rule combines them along the tape into the derivative of the NLL with respect to every parameter. In plain English, RTMB differentiates the code step by step, as we would differentiate a formula by hand. Mention it only if asked; the word tape is not on the slide.

Break (10 minutes).

Slide 13 — Precision and correlation of curve parameters

The uncertainty of the estimates is calculated from the likelihood, as in Chapter 4, and `sdreport()` gives the standard errors and the covariance matrix from the curvature of the NLL. Two features are typical of the parameters of a curve. Their estimates are often strongly correlated, because different combinations of the parameters give similar curves over the range of the data. And how precisely each parameter is estimated depends on where along x the observations lie, which is the subject of slide 14. The correlation also depends on the parameterisation, whereas the fit does not, as Chapter 4 showed for the owl model in two parameterisations.

For the tanks, the standard errors on the optimised scales are 0.285 for logit $(\hat{\alpha})$ and 0.294 for $\log(\hat{H})$, and the correlation between the two estimates is 0.82. It is positive because a steeper initial rise of the curve (a larger α) must be compensated by a stronger levelling off (a larger H) for the curve to pass through the tanks at high density, so hypothetical samples that overestimate one parameter tend to overestimate the other. In the Michaelis–Menten form, with $\log(a)$ and $\log(b)$, the same model has the same maximum and a correlation of 0.98: the estimates of the asymptote

and of the half-maximum are almost interchangeable, because the data determine mainly their ratio, the initial slope $a/b = \alpha$. This is what Chapter 5 found when it read starting values from these data.

The two standard errors are similar on the optimised scales, but that does not make the two parameters equally well determined: the next two slides show that the tanks determine α much better than H.

Delivery

Show the chunk without running it. The fit in the Michaelis–Menten form is a hidden chunk of the chapter; quote its correlation without showing code.

Slide 14 — Where the data lie

Chapter 5 showed with partial derivatives that a curve responds to each parameter in a different part of its range. A **partial derivative** with respect to a parameter says how much the curve moves at each x when that parameter changes and the others stay fixed. The observations that inform a parameter are those made where the curve responds to it. So data that cover only the rise of a saturating curve determine its initial slope better than its asymptote, and the design of a study (where along x the observations are made) determines what can be estimated.

The densities of the reed frog experiment stop at 100 tadpoles per tank, below the estimated half-maximum of 115, so the tanks cover less than half of the rise of the curve. They determine α, which describes the start of the curve, much better than H, which describes its end.

Figure walkthrough

The figure is Chapter 5's figure of the two partial derivatives of $ax/(b + x)$ with $a = 60$ and $b = 120$, the values read by eye, for x from 0 to 600. On the left, the derivative with respect to the asymptote a rises from 0 towards 1: a change in a moves the curve most at large x, where the curve is close to its asymptote. It is 0.5 at $x = 120$. On the right, the derivative with respect to the half-maximum b is negative (a larger half-maximum means a lower curve), zero at both ends and largest in size at $x = b$ (the dashed line), where it is $-a/(4b) = -0.125$. Point at the part of the horizontal axis between 5 and 100, where the tanks lie: there the derivative with respect to a is still below one half, and both derivatives grow in size together, so a change in one parameter can largely be compensated by a change in the other. That is the correlation of 0.98 of slide 13. The axis labels of this copy are small at slide size; read the values aloud.

Transition

The intervals of the next slide put numbers on "well determined" and "poorly determined".

Slide 15 — Likelihood profile intervals of curve parameters

Likelihood profile intervals are calculated with `TMB::tmbprofile()` and `confint()`, as in Chapter 4, and they remain the reference method. With a response curve, the profile of a poorly determined parameter is often asymmetric, because the data rule out values on one side of the estimate much more strongly than on the other. The endpoints of the interval can be transformed with any

monotonic function to give an interval for a quantity calculated from the parameter: an increasing function keeps their order and a decreasing function swaps them. The coverage is kept, because the transformed interval contains the transformed quantity in exactly the hypothetical samples in which the original interval contains the parameter.

For the reed frogs, the interval for the attack rate runs from 0.40 to 0.68 per experiment, which is narrow compared with the interval for the handling time, 0.0070 to 0.026 experiments per tadpole killed, almost a factor of four. The largest number of tadpoles that the larvae could kill in one experiment, $1/H$, decreases with H, so the endpoints swap: its interval runs from 38 to 143 around an estimate of 60. It extends 22 tadpoles below the estimate and 83 above it, and its upper end lies beyond the largest density of the experiment. The data rule out a low asymptote, because the larvae killed up to 35 tadpoles in one tank, but not a high one. The profile is asymmetric even on the log scale, so the Wald interval from the log scale, 34 to 107 tadpoles, misses most of the upper part of the profile interval.

Delivery

Show the chunk without running it; the demo draws the profile of `log_H`. `confint()` warns "collapsing to unique 'x' values", because each profile contains the estimate twice; the warning can be ignored, and the chapter's chunk suppresses it with a comment.

Slide 16 — The uncertainty of a fitted curve

A point of the curve, such as the expected number killed at a given density, is a quantity calculated from all the parameters. Its standard error is requested with `ADREPORT()`, as in Chapter 4, and TMB applies the delta method. `ADREPORT()` of a vector reports every element with its own standard error. So if we pass a grid of predictor values in the list of data and calculate the curve on that grid inside the NLL, `sdreport()` returns the curve and its standard error at every point of the grid. From them we calculate a Wald interval at each point, on a scale on which the quantity is unbounded, and transform the endpoints back, for the same reasons as for the parameters in Chapter 4: the interval cannot cross a bound, and the log-likelihood is usually closer to a parabola on that scale. For a probability, that scale is the logit. Together the intervals form a **confidence band**.

The band is **pointwise**: at each value of x, the interval contains the value of the curve under the data generating process at that x in approximately 95% of hypothetical samples. A band that contained the whole curve in 95% of hypothetical samples would be wider.

For the reed frogs, the grid holds the densities from 0 to 100 tadpoles per tank, as `initial_grid` in the list of data, and the NLL gains three lines: the probability that a tadpole is killed at every density of the grid, its logit with `qlogis()`, and `ADREPORT()` of that vector. We process the new NLL and fit it again (the estimates do not change), and `summary()` of `sdreport()` with `select = "report"` returns one row per density of the grid. The Wald interval of each row is calculated on the logit scale, its endpoints are transformed back with `plogis()` and multiplied by the density to give the band for the expected number killed.

Whether these intervals have a coverage close to 95% with 16 tanks can be checked by simulation, in the same way as in "Coverage checked by simulation" in Chapter 4. The chapter does not do it; the extra exercise of the practical does it for a quantity of a fitted curve.

Optional remark (the box of the chapter)

The statistical detail box gives the delta method for several parameters; do not write the formula on the board unless asked. Close to the estimates, the quantity g is replaced by its tangent plane. Each parameter contributes its variance multiplied by the square of the slope of g in its direction, and each pair of parameters contributes its covariance multiplied by the two slopes. For the reed frog curve a larger α raises the curve and a larger H lowers it, so the two slopes have opposite signs, while the estimates are positively correlated. The covariance term is therefore negative: the two errors partly cancel on the curve, which is estimated more precisely than the separate standard errors suggest. The approximation is poor when the quantity is close to a bound of its range on the scale on which it is calculated, which is another reason for the logit scale.

Slide 17 — A confidence band

A confidence band is read point by point: at each value of the predictor, the shaded interval is the 95% confidence interval of the expected response at that value. It describes the uncertainty of the curve and says nothing about where the observations of new tanks would fall. That is described by the distribution of the observations around the curve (slide 5), which is much wider, so many observations lie outside a confidence band and this is no sign that something is wrong. Beyond the range of the data the band widens quickly, because there the curve depends more and more on the parameters that the data determine poorly.

For the reed frogs, on the logit scale the band is narrowest near 50 tadpoles per tank and wider towards both ends. The tanks at low density hold few tadpoles each, so most of the information about the curve comes from the denser tanks, and the band widens away from the middle of that information. For the expected number killed the band also widens with density, because the probability is multiplied by the number of tadpoles. At 100 tadpoles per tank it runs from 23.5 to 33.1 tadpoles killed, around an expected 28. Within the range of the data the curve is well determined, although its asymptote is not.

Figure walkthrough

The blue points are the 16 tanks, the vermillion line is the fitted expected number killed, and the light vermillion shading is the pointwise 95% band, calculated from Wald intervals on the logit scale. Point out how many tanks lie outside the band, and ask why (the answer is the second bullet). The inset gives the model, the estimates and their likelihood profile intervals: α is per experiment, H in experiments per tadpole killed and $1/H$ in tadpoles killed per experiment. A figure that reports a fitted response should show these together: data, curve, uncertainty, the model and the estimates with their intervals.

Slide 18 — Practice 2: handling time of a ladybird

Short activity — allow about two minutes of work and one to discuss

Students use R, or a calculator. Expected answers, from the chapter's solution:

1. The standard error belongs to $\log(\hat{H})$, so the interval is calculated on the log scale and its endpoints transformed with `exp()`: `H_wald = exp(-3.2 + c(-1, 1) * qnorm(p = 0.975) * 0.20)` gives approximately 0.028 to 0.060 hours per aphid.
2. The inverse of the two endpoints, in reverse order, `rev(1 / H_wald)`: approximately 17 to 36 aphids/hour, around an estimate of $1/e^{-3.2} \approx 24.5$.
3. The function $1/H$ decreases as H increases, so the lower endpoint of H gives the upper endpoint of $1/H$, and the other way round. Because the function is monotonic, the second interval contains $1/H$ in exactly the hypothetical samples in which the first contains H, so it has the same coverage.

A common slip is to take `exp(0.20)` as a standard error of $\hat{H}$; recall Chapter 4: `exp()` of a standard error is no standard error at all.

Slide 19 — Simulating from a fitted model

A fitted model describes the whole distribution of every observation, and not only its expected value. **Checking** the model means comparing that description with the data. In Chapter 3 the observations had a common distribution, so we could compare their empirical distribution with the model distribution. With a response curve each observation has a distribution of its own, and only one observation comes from each of them. We therefore simulate many values from the distribution of each observation and see where the observed value falls among them.

With its parameters at the estimates, a fitted model is a data generating process from which we can simulate. As in Chapter 4, we mark the response with `OBS()` inside the NLL, and the function `simulate()` of the processed object then draws one hypothetical sample for the parameter values that we give it. A hypothetical sample has the same design as the data: the same observations, with the same values of the predictor. Each simulated value is drawn from the distribution of its observation, so the simulation contains the variation of the observations around the curve, which is what the check needs.

For the reed frogs, the NLL gains one line, `killed = OBS(killed)`, and we process and fit it again (the estimates do not change). Each call to `simulate()` returns the numbers killed in 16 simulated tanks with the initial numbers of the experiment, and the loop stores 1,000 hypothetical samples in a matrix with one row per tank and one column per hypothetical sample.

Delivery

Show the chunk without running it; the demo runs it. Seed 601 gives the simulations of the chapter, and so the residuals of slides 20 and 21.

Slide 20 — Scaled quantile residuals

The **scaled quantile residual** of an observation is the fraction of its simulated values that are smaller than the observed value. An observation in the middle of its simulated values has a residual close to 0.5, one below all of them a residual of 0, and one above all of them a residual of 1. For a discrete response, the observed value is often equal to some of the simulated values, and the

residual is then drawn at random between the fraction of simulated values that are smaller and the fraction that are smaller or equal.

If the model describes the data generating process, each observation behaves like one more draw from its own distribution, so it is equally likely to fall anywhere among its simulated values. The residuals are then approximately uniform between 0 and 1, whatever the distribution of the observations and whatever the values of the predictor. This is what makes them useful for response curves: they put observations with different distributions on one common scale.

The package DHARMa calculates them. `DHARMa::createDHARMa()` takes the matrix of simulated values (one row per observation, `simulatedResponse`), the observed values (`observedResponse`), the expected value of each observation under the fitted model (`fittedPredictedResponse`, which DHARMa uses in its plots and in the dispersion statistic of slide 21), and `integerResponse = TRUE` for a discrete response, so that ties are drawn at random as described above.

For the reed frogs, one tank, with 50 tadpoles of which 5 were killed, lies below all its 1,000 simulated values, so its residual is 0.

Optional remark (citations)

Dunn and Smyth (1996), "Randomized quantile residuals", *Journal of Computational and Graphical Statistics* 5: 236–244, defined these residuals from the cumulative probability function of the model and transformed them to the scale of a standard Normal distribution. DHARMa (Hartig, R package version 0.5.0) calculates them from simulations and keeps them between 0 and 1.

Slide 21 — Reading residual plots

Three kinds of pattern show that the model disagrees with the data. A distribution of the residuals that is not uniform: too many residuals near 0 and 1 mean that the observations vary more around the curve than the model allows, and too many near 0.5 that they vary less. A trend of the residuals against the predictor or against the fitted values, which means that the curve misses a systematic change of the response. And residuals of exactly 0 or 1, from observations outside the range of all their simulated values, which DHARMa marks as outliers.

`plot()` of the result of `createDHARMa()` draws two panels: on the left a quantile–quantile plot of the residuals against a uniform distribution, and on the right the residuals against the fitted values, on the scale of their ranks, with the quartiles 0.25, 0.5 and 0.75 as dashed lines. `DHARMa::plotResiduals()` with the argument `form` draws the residuals against any predictor. For the reed frogs the expected number killed increases with the initial number of tadpoles, so the right panel already shows the residuals in the order of the densities.

DHARMa summarises the spread with a **dispersion statistic**: the variance of the observed values around their expected values, divided by the average of the same variance in the simulated samples. It is close to one when the model allows the right amount of variation. For the reed frogs it is 2.9: the numbers killed vary around the curve almost three times as much (as a variance) as the Binomial model allows. This is **overdispersion** relative to the Binomial distribution, which Chapter 2 introduced for counts relative to the Poisson distribution. The two tanks with 50 tadpoles, with 5 and 21 killed, show it in the data: under the Binomial model they share the same

distribution, in which 5 killed is very unlikely. The model assumes that every tadpole of a tank has the same probability of being killed, independently of the other tadpoles, and these tanks differ more than that assumption allows.

DHARMa also prints tests with p-values, on its plots and in functions such as `testDispersion()`. We read the plots and the dispersion statistic as exploratory evidence and do not rest decisions on the p-values, for the reasons given in Chapter 1 and at the end of Chapter 4: with 16 tanks a large p-value mainly says that there are few data, and a small one does not say how much the disagreement matters.

The takeaway is the chapter's warning: a check shows where a fitted model disagrees with the data, not which assumption is wrong. Overdispersion can come from the distribution, from a predictor that is missing from the curve, or from observations that are not independent. Nor does a plot without a pattern show that the model is right; with few observations only large disagreements are visible.

Figure walkthrough

The figure is DHARMa's own plot, as the chapter draws it. On the left, the lower points lie close to the diagonal and the upper half of the residuals lies above it: more tanks than expected have residuals close to 1. The star on the right panel, at zero, is the tank with 5 of 50 killed. On the right, the lines fitted to the quartiles of the residuals lie on or outside the dashed lines, the upper one well above 0.75: the tanks spread further from the middle of their simulated values than the model allows. There is no clear trend against density, so the curve itself does not miss a systematic change. The printed tests are on the figure because DHARMa draws them; point at the dispersion statistic in the example box instead.

Slide 22 — Discussion: what does a residual pattern tell us?

This discussion applies the three kinds of pattern and the warning of slide 21.

Discussion and expected answers — included in the three minutes

Give students two minutes to label the four patterns with a neighbour, then take one answer per item.

1. **Curve:** the observations are lower than the model predicts at low x and higher at high x, so the curve misses a systematic change: its shape is wrong, or a predictor that changes with x is missing. It says nothing in particular about the distribution (quiz question 6 of the chapter).
2. **Distribution**, with the caveat of slide 21: the observations vary more than the model allows at every x, which a distribution with more variation (slide 24) may describe. A missing predictor or a lack of independence can produce the same pattern, so accept "cannot tell" if the student says why.
3. **Independence:** residuals of neighbouring quadrats that are alike mean that the observations share something that the model does not contain. Chapter 7 deals with observations that come in groups.
4. **Cannot tell:** with eight observations only large disagreements are visible, and a plot without a pattern does not show that the model is right.

Break (10 minutes).

Slide 23 — Live demo: the modelling cycle for one model

The demo runs the steps of slides 8 to 21 for one model, the Binomial model with the Holling type II curve, from the fit to the residuals, with the script `demo-cycle.R`. It has no model comparison: the comparison follows on slides 26 to 29. The script needs the packages RTMB, RTMBdist, TMB (for `tmbprofile()`), `emdbook` (for the data) and DHARMa. Its code and object names are those of the chapter, and the simulation uses the chapter's seed (601), so the numbers and the residual plot agree with the chapter and the slides. The whole script runs in a few seconds (about 6 seconds in a check in R 4.6.1), so there is no waiting.

During the break, open R and run section 0, "Packages and data". It loads RTMB and RTMBdist (with their messages about masked functions), prints the 16 tanks, builds the list of data `frog_data` and defines `cmb()`. Keep the session open. Spend the two minutes at the slide reading the five steps, then switch to R and run the sections one at a time.

Walkthrough of the script — 16 minutes

1. **The NLL and the fit (about 4 minutes).** Define the NLL and call it directly with two lists, the values by eye and the data: it returns 46.95. This is the check of slide 12, before processing. Then process it and print the gradient at the starting values, -4.30 and 3.20: ask which way each parameter must move (up for logit (α), down for $\log(H)$). Fit, and show the convergence code 0 and the gradient at the estimates (about 0.002 and -0.005, close to zero).
2. **Estimates and the curve (about 2 minutes).** $\hat{\alpha} = 0.526$ per experiment, $\hat{H} = 0.0166$ experiments per tadpole killed, an asymptote of 60.2 tadpoles killed and a half-maximum of 114.6 tadpoles per tank. The plot draws the tanks, the fitted curve (solid, vermillion) and the curve by eye (dashed). Point out that the two curves nearly coincide and that both are still rising at 100 tadpoles per tank.
3. **Uncertainty of the parameters (about 3 minutes).** `summary()` of `sdreport()`: 0.104 with a standard error of 0.285 for `logit_alpha` and -4.098 with 0.294 for `log_H`; `cov2cor()` gives 0.82. The plot of the profile of `log_H` shows its asymmetry: it falls more slowly towards low values of H, that is, towards a high asymptote $1/H$, which the data rule out less strongly than a low one. `confint()` gives 0.0070 to 0.0264 for H and, with the endpoints swapped, 37.9 to 143.1 for $1/H$. R prints the warning "collapsing to unique 'x' values" twice; it can be ignored (slide 15).
4. **Uncertainty of the curve (about 3 minutes).** The grid of densities from 0 to 100 goes into the list of data, and the NLL gains `OBS()` (used in section 5; it does not change the fit) and the three lines of `ADREPORT()`. After the new fit, `head(band_summary)` shows one row per density; ask why the first row equals the row of `logit_alpha` of section 3 (0.104 and 0.285): at a density of zero, $p(0) = \alpha$. The band at 100 tadpoles per tank runs from 23.5 to 33.1 around 28.1 killed. The plot draws the band as dashed lines.
5. **Simulation and residuals (about 4 minutes).** One hypothetical sample first: the numbers killed in 16 simulated tanks with the densities of the data. It is drawn before the seed is set, so its values differ at every run; that is the point (same design, different values). Then the 1,000

samples with seed 601, the expected values and `createDHARMa()`; `plot()` draws the figure of slide 21. The table of residuals shows the tank with 5 of 50 killed at 0.000, and six of the 16 tanks below 0.1 or above 0.9 (0.000, 0.072, 0.078, 0.921, 0.961 and 0.978), where about three would be expected if the residuals were uniform. The dispersion statistic is 2.93. End by saying that this is where the cycle sends us back to the model: slide 24.

Questions to ask during the demo (about 1 minute, within the sections above)

1. In section 1, "What would happen if the NLL contained `if (H > 0.02)`?" `MakeADFun()` would stop with an error, because RTMB refuses a comparison of a parameter with a value (slide 12). Do not run it.
2. In section 4, "Do 95% of the tanks lie inside the band?" No: the band is for the expected number killed, and the tanks vary around it according to the Binomial distribution (slide 17).
3. In section 5, "Is the tank with 5 of 50 killed an error in the data?" The residuals cannot say: it is very unlikely under this model, which is what overdispersion means, and a model in which the probability varies among tanks may allow it (slide 24).

Slide 24 — Revising the observation model

When the probability of an outcome varies among sampling units beyond what the predictor explains, the counts vary more than a Binomial model allows. The **Beta-Binomial** distribution describes this situation: the probability of each sampling unit is drawn from a Beta distribution, and the count of that unit follows a Binomial distribution with this probability. With a response curve we parameterise it with the mean probability p, which the curve gives, and the **precision** κ (in the supplement $\kappa = a + b$, the sum of the two shape parameters of the Beta distribution, so $a = \kappa p$ and $b = \kappa(1 - p)$).

For a unit with N trials, the mean is Np, as for the Binomial distribution, and the variance is the Binomial variance $Np(1 - p)$ multiplied by the factor $(\kappa + N)/(\kappa + 1)$. The factor is larger than one whenever $N > 1$ and approaches one as κ grows, so a small κ means much variation among units and a large κ gives almost a Binomial distribution. Note that the factor grows with N: the variation of the probability among units matters more when each unit has more trials. The precision is positive, so we optimise $\log(\kappa)$.

The curve does not change: only the observation model is revised, so that the probability that a tadpole is killed varies among tanks around $p(N)$. This is the step of the modelling cycle in which a check sends us back to the model. The NLL uses `dbetabinom()` of RTMBdist, with the shape parameters κp and $\kappa(1 - p)$, and includes `OBS()` from the start, because this model is checked in the same way. The fit starts from the estimates of the Binomial model and a precision of 10.

The estimates of the curve hardly change: $\hat{\alpha} \approx 0.54$ per experiment and $\hat{H} \approx 0.018$ experiments per tadpole killed. Their standard errors on the optimised scales are larger, 0.38 against 0.29 for logit $(\hat{\alpha})$ and 0.46 against 0.29 for $\log(\hat{H})$. The Binomial model treated every tadpole as an independent trial, so it took the tanks as more informative than they are. When the probability varies among tanks, the tadpoles of a tank share part of their fate, and the information about the curve depends more on the number of tanks than on the number of tadpoles; that is the takeaway.

The estimated precision, $\hat{\kappa} \approx 26$, gives a variance factor of about 1.3 for a tank with 10 tadpoles and about 4.6 for a tank with 100. The check is repeated with the new object: the dispersion statistic is now 0.92, and the tank with 5 of 50 killed lies within the range of its simulated values. The Beta-Binomial model allows about as much variation among tanks as the data show.

Delivery

The code is on the chapter page ("A model for overdispersed proportions"); it is not run live. If a student asks, the Beta-Binomial entry of the distribution supplement gives the other parameterisation.

Slide 25 — Practice 3: fruit set in inflorescences

Short activity — allow about two minutes of work and one to discuss

Students work on paper. Expected answers, from the chapter's solution:

1. The Binomial variance is $20 \times 0.4 \times 0.6 = 4.8$. With $\kappa = 5$ the factor is $(5 + 20)/(5 + 1) \approx 4.17$, so the Beta-Binomial variance is $4.8 \times 25/6 = 20$, more than four times larger.
2. For a variance 10% larger the factor must be 1.1: $(\kappa + 20)/(\kappa + 1) = 1.1$ gives $\kappa + 20 = 1.1\kappa + 1.1$, so $\kappa = 189$.

The chapter's callout also asks students to verify the variance for $\kappa = 5$ in R from the probability mass; they can do that at home. The second answer shows how large κ must be before a Beta-Binomial model with 20 trials per unit is close to a Binomial one.

Slide 26 — Candidate models

The candidate models of a comparison should represent plausible ecological or statistical alternatives, and they must all be fitted to the same observations: the same rows, the same response and the same units. Often some candidates are nested in others: a model is **nested** in another when it is the special case obtained by fixing a parameter of the other at a particular value, as for the curves of Chapter 5 ("Nested functions").

For the reed frogs we have two questions. Does the probability that a tadpole is killed decrease with density? And do the tanks vary more than a Binomial model allows? Each question gives two alternatives, and together they give four candidates: two curves crossed with two distributions. The constant probability is the Holling type II curve with $H = 0$: the larvae kill the same fraction of the tadpoles at every density, so the expected number killed, αN, increases in proportion to the density. The Binomial distribution is the limit of the Beta-Binomial distribution as κ grows without bound. So the model with a constant probability and a Binomial distribution is nested in each of the others, and each Binomial model is nested in the Beta-Binomial model with the same curve. The Holling type II curve with a Binomial distribution and the constant probability with a Beta-Binomial distribution are not nested in each other, but they can still be compared, because they are fitted to the same observations.

Delivery

The special values $H = 0$ and $\kappa = \infty$ cannot be reached on the optimised scales (they would be $\log(H) = -\infty$ and $\log(\kappa) = \infty$), so the two models with a constant probability have NLLs of their own in the chapter. The chapter fits all four with a small function, `fit_model()`, and a loop; all four fits converged. The maximum log-likelihood is highest for the model with the most parameters, as it must be for nested models, which is why the next slides need criteria that take the number of parameters into account.

Slide 27 — AIC, AICc and BIC

Chapter 4 introduced AIC and its corrected version AICc, where K is the number of estimated parameters, n the number of observations and $\ell(\hat{\theta})$ the maximum log-likelihood; do not repeat its derivation. With 16 observations and up to three parameters the correction of AICc is not negligible: it adds 2 units to the model with three parameters, against about 0.3 for the model with one.

The **Bayesian information criterion** (BIC) has a penalty that grows with the number of observations: each parameter costs $\log(n)$ units instead of two, which is more than two when n is larger than $e^2 \approx 7.4$ (for 16 tanks, $\log(16) \approx 2.8$). So for most data sets BIC penalises additional parameters more strongly than AIC. The two criteria also have different aims. AIC estimates how well the fitted model would predict new data from the same process. BIC tends, as the sample grows, to select the model with the fewest parameters among the candidates that contain the data generating process. So in large samples BIC tends to prefer simpler models than AIC does. Like AIC, BIC is used as a difference between models fitted to the same observations, the lowest value is preferred, and it ranks the candidates without showing that the best of them describes the data well.

In AICc and BIC, n is the number of independent observations, which is the number of tanks, 16, and not the 610 tadpoles that they contained: under the Beta-Binomial model the tadpoles of a tank share the probability of that tank, so they are not independent observations. The same n must be used for every model that is compared.

Delivery

If a student asks whether n could be 610 for the two Binomial models, say that the same n must be used for all the models that are compared, and that Chapter 7 returns to the question of what counts as an independent observation.

Optional remark (citation)

Schwarz (1978), "Estimating the dimension of a model", *Annals of Statistics* 6: 461–464. The name comes from its derivation, as a large-sample approximation in Bayesian inference, but it is used in the same way as AIC.

Slide 28 — Leave-one-out cross-validation

Cross-validation estimates how well a model predicts observations that were not used to fit it. The observations are split into parts, the model is fitted to all but one part and scored on the part that was held out, and this is repeated until every part has been held out once. In **leave-one-out**

cross-validation each part is a single observation: every parameter of the model is estimated again from the other $n - 1$ observations, and the held-out observation is scored by its **log score**, the log probability (or log density) that the refitted model assigns to it. The score of the model is the sum of the log scores of the n observations, and a larger (less negative) score is better.

The log score uses the whole distribution of the observation, and not only its distance from the curve, so it can be calculated for any observation model, and it rewards a model that describes the spread of the observations as well as their mean. A model that adapts too closely to the data used to fit it predicts the held-out observations badly, so cross-validation needs no penalty for the number of parameters. Leave-one-out cross-validation and AIC agree in large samples: AIC is approximately -2 times the leave-one-out log score. When fitting the model n times takes too long, the observations are split into a few groups that are held out in turn, often 5 or 10 (5-fold or 10-fold cross-validation). When the observations are ordered in time or space, so that neighbouring observations are similar, each held-out part should be a block of neighbouring observations; otherwise its neighbours in the fit make it look easier to predict than a new observation would be.

For the reed frogs we leave out one tank at a time: 16 fits per model, 64 in all, each from the same starting values as the fit to all the tanks, and all of them converged. Every parameter is refitted, also κ; fixing a parameter at its estimate from all the tanks would let the held-out tank contribute to its own prediction (quiz question 10 of the chapter).

Delivery

In the chapter, the log score of the held-out tank is calculated by calling the NLL directly with two lists: the estimates from the other 15 tanks (with `as.list()`) and the data of the held-out tank. Minus that NLL is the log probability of the tank. Leave-one-out needs no random split and no seed: every tank is held out once.

Optional remark (citation)

Stone (1977), "An asymptotic equivalence of choice of model by cross-validation and Akaike's criterion", *Journal of the Royal Statistical Society, Series B* 39: 44–47.

Slide 29 — Comparing the candidates

Only differences between models matter for each criterion, so the table gives the difference of each model from the best one: the lowest AIC, AICc and BIC, and the highest leave-one-out score, whose differences are therefore zero or negative. Recall that a difference of one unit in the log score corresponds to about two units of AIC.

For the variation among tanks the criteria agree: with either curve, every criterion prefers the Beta-Binomial model to the Binomial model, by 6.7 (Holling type II) and 13.0 (constant probability) units of AIC, and by 3.3 and 8.2 units of log score. This agrees with the residuals of slide 21.

For the density dependence of the risk the answer depends on the distribution. Under the Binomial models, the Holling type II curve is clearly better than a constant probability (9.0 units of AIC, 4.0 of log score). Under the Beta-Binomial models the difference is small, and the criteria do not agree on its sign: AIC, AICc and BIC prefer the Holling type II curve by 2.7, 1.6 and 2.0 units, whereas the

leave-one-out score prefers the constant probability by 0.9 units. The Binomial models overstated the evidence for density dependence for the same reason that they understated the standard errors: they treated the tadpoles as independent trials. With 16 tanks that vary among themselves, the data say clearly that a tadpole is killed with a probability of about one half at low density and that the tanks vary more than a Binomial model allows, but they hardly distinguish a probability that decreases with density from a constant one over densities of 5 to 100 tadpoles per tank.

For the ecological answer the chapter uses the Beta-Binomial model with the Holling type II curve, which three of the four criteria prefer and which contains the constant probability as the case $H = 0$. Its attack rate is estimated at 0.54 per experiment, with a likelihood profile interval from 0.38 to 0.76: at low density, between about two fifths and three quarters of the tadpoles are killed on average in the 14 days. Its handling time is estimated at 0.018 experiments per tadpole killed, with an interval from 0.0019 to 0.038, so the largest number of tadpoles that three larvae could kill in one experiment, about 56, has an interval from 27 to more than 500. The lower end of the interval of H is close to zero, which is the constant probability, and this agrees with the comparison. To estimate the handling time, an experiment would need tanks with densities well above the half-maximum, where the curve levels off (slide 14).

Delivery

Read the table by columns, then by rows: the two rows of the Binomial models are far from the best for every criterion; the two rows of the Beta-Binomial models are close, and the last column has the opposite sign. The takeaway is the general point: the criteria rank the candidates and can disagree, and what the ranking means for the question is a matter of ecological interpretation. Students should see that model comparison criteria can disagree; do not present the choice of the last model as settled by the criteria.

Slide 30 — A recipe for a response curve

The steps of this chapter extend the recipe of Chapter 4 to models in which a curve describes a parameter of the distribution of each observation, and the chapter collects them in a note callout before the summary. Read the ten steps with the tool of each: choose the curve and the observation model, with the range that the parameter requires; write the NLL with the predictor and the response in the list of data; read starting values from a plot; fit, check `convergence` and plot the curve over the data; standard errors and correlations with `sdreport()`, likelihood profile intervals with `TMB::tmbprofile()` and `confint()`; the curve on a grid with `ADREPORT()` and its band; `OBS()`, `obj$simulate()` and the scaled quantile residuals; revise the curve or the observation model and check the revised model in the same way; compare the candidates with AIC, AICc, BIC and leave-one-out cross-validation; and interpret the estimates and intervals of the preferred model in units, in terms of the ecological question, saying what the data leave uncertain.

Delivery

About two minutes. Steps 2, 4 and 5 are those of Chapter 4; point at the new ones (1, 3, 6 to 8 and 10). The practical follows the same sequence.

Slide 31 — The Chapter 6 R practical

The practical fits response curves to three data sets that the Chapter 5 practical used as a backdrop for curves placed by eye. Exercises 1 to 3 are the core and Exercise 4 is extra practice. Exercise 1, on the titre of the myxoma virus in infected rabbits (`emdbook`), is worked on the page: a Ricker curve with a Normal observation model, Wald intervals of the time and the height of the peak from `ADREPORT()`, the likelihood profile interval of the time of the peak, residuals with DHARMa, and a Gamma observation model compared with AIC, AICc and BIC. Exercise 2, on the rate of an enzymatic reaction (`Puromycin`), fits a Michaelis–Menten curve with one asymptote per state (cells treated with puromycin or not), the difference between the asymptotes from `ADREPORT()`, and a comparison with information criteria and leave-one-out cross-validation. Exercise 3, on the fecundity of fir trees (`emdbook`), fits a power law with Poisson and Negative Binomial models and asks whether the data support a fecundity proportional to the cross-sectional area of the trunk. Exercise 4 simulates the coverage of the Wald interval of the time of the peak of Exercise 1, with a skeleton function that students complete.

Try each exercise before checking the solution.

Slide 32 — Summary and closing questions

Go back to slide 3 by speaking about it again; no new slide is needed. A response curve describes one parameter of the distribution of each observation, usually its mean or, for a Binomial distribution, its probability, and it must return values in the range of that parameter. The observations are treated as independent, but each has its own distribution. The log-likelihood is the sum of the log probabilities of the observations, each at its own value of the predictor, and it is maximised with the tools of Chapter 4, with RTMB supplying the exact gradient. The parameters of a curve are often strongly correlated, and their precision depends on where the observations lie along the predictor. `ADREPORT()` on a grid gives a pointwise confidence band, calculated on an unbounded scale. Simulated residuals show where a fitted model disagrees with the data, but not why; for the reed frogs they showed overdispersion, and a Beta-Binomial model with the same curve described the variation among tanks. Candidate models fitted to the same observations are compared with AIC, AICc, BIC and leave-one-out cross-validation, with the number of independent observations as n. On the diagram of slide 3, the lecture went once round the whole cycle: model, estimation, check, revision, comparison and answer.

Return to this slide for the two closing questions.

Discussion and expected answers — allow about two minutes

Question 1: observations inform a parameter where the curve responds to it. On the rise of a saturating curve the curve responds mainly to its initial slope; the asymptote matters only where the curve levels off. So many combinations of asymptote and half-maximum with the same initial slope give similar curves over the data: the estimates are strongly correlated and the interval of the asymptote is wide and asymmetric (slides 13 to 15).

Question 2: which assumption is wrong. A pattern can come from the distribution, from a predictor that is missing from the curve or from observations that are not independent; and a

plot without a pattern does not show that the model is right, especially with few observations (slide 21).

Final transition

In this chapter every tank was an independent observation. In Chapter 7 the observations come in groups, such as leaves of the same plant, visits to the same nest or plots at the same site, which are not independent of each other, and the parameters of the curve vary among groups. The practical comes first.

Adjustments during delivery

The slides take 121 minutes and leave 9 minutes of slack, which comes last. The demo is kept whole; if it overruns, it takes the slack first. If discussion overruns, use the slack first, then shorten in this order:

- Slide 3 to one sentence: recovers about half a minute.
- Slide 4 to the two columns and the takeaway, with the provenance in one sentence: about one minute.
- Slide 11 to the figure and the example box: about one minute.
- Slide 14 to the takeaway, pointing at the figure once: about two minutes.
- Slide 22 to two of the four patterns (1 and 2): about one and a half minutes.
- Slide 25 given as homework, with the answers in one sentence: about two and a half minutes.
- Slides 30 and 31 to one sentence each: about two minutes together.

Keep the response curve as a parameter of a distribution and its range (slides 5 and 6), the likelihood of a response curve (slide 8), the correlation and precision of curve parameters (slide 13), the uncertainty of the curve (slide 16), the definition of the scaled quantile residuals (slide 20) and the revision of the observation model (slide 24).

The breaks are placed between the blocks, after slides 12 and 22 (minutes 36 and 83). If a block runs over, shorten the block rather than the following break.

If time is left: code to run

No code is run live apart from the demo, which stops before the comparison. Students can follow the chunks of the chapter page at home, by section: “A model for overdispersed proportions” (the Beta-Binomial NLL with `dbetabinom()`, its fit, its simulations and its dispersion statistic), “Comparing candidate models” (the two NLLs with a constant probability, `fit_model()`, the list `models` and the loop that fits the four candidates), “Information criteria” (AIC, AICc and BIC with `n_tanks`), “Cross-validation” (`subset_tanks()` and the loop that leaves out one tank at a time) and “Comparing the four models” (the table of differences and the likelihood profile intervals of the Beta-Binomial model).